

1. Process Scheduling:

- (a) (5 points) Explain the difference between waiting time and turnaround time. These times are both used to measure the effectiveness of scheduling schemes.
每個 process wait 的時間. 每個 process / turn.
- (b) (5 points) Which of the following is true of multilevel queue scheduling?
- A) Processes can move between queues.
 - B* B) Each queue has its own scheduling algorithm.
 - C) It is the most general CPU-scheduling algorithm.
 - D) None of the above.
- (c) (10 points) Explain the process of starvation and how can we prevent it.
時間的 req.

2. Synchronization:

- (a) (15 points) In the first readers-writers problem, readers have higher priorities to access a shared object (i.e., file). Design the entry and exit sections of the reader and writer processes using the following (shared) data structures:

```
semaphore mutex, wrt; // initialized to 1
int readcount; // initialized to 0
```

// reader's protocol

```
do {
    if (readcount == 1)
        wait (mutex);

    // reading is performed
    signal (mutex);
    if (readcount == 0)
        signal (wrt);
} while (TRUE);
```

// writer's protocol

```
do {
    wait (wrt);

    // writing is performed
    signal (wrt);
} while (TRUE);
```


- (b) (5 points) A counting semaphore ____.
- A) is essentially an integer variable
 - B) is accessed through only one standard operation
 - ☒ C) can be modified simultaneously by multiple threads
 - D) cannot be used to control access to a thread's critical sections

Unsafe
Safe

什麼情況 deadlock
什麼情況 不 deadlock

3. Deadlocks:

- (a) (5 points) Explain why an unsafe state MAY BUT NOT NECESSARILY lead to a deadlock.
- (b) (10 points) A method to prevent deadlock is to impose a total ordering of all resource types. And each process can request resources only in an increasing order of enumeration. Proof that the circular-wait condition cannot hold if the method is employed. $P_1 \leftarrow P_2 \leftarrow P_3 \dots \leftarrow P_1$
- (c) (5 points) Which of the following statements is true?
- ☒ A) A safe state is a deadlocked state.
 - ☒ B) A safe state may lead to a deadlocked state.
 - ☒ C) An unsafe state is necessarily, and by definition, always a deadlocked state.
 - D) An unsafe state may lead to a deadlocked state.

4. Memory Management Strategies:

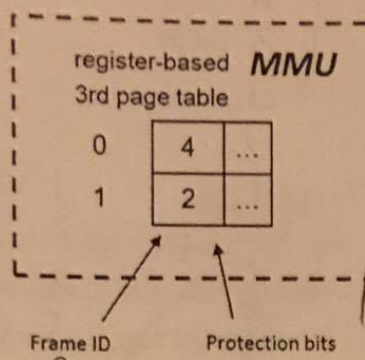
- (a) (5 points) Which of the following is true of compaction? with
- A) It can be done at assembly, load, or execution time.
 - B) It is used to solve the problem of internal fragmentation.
 - C) It is possible only if relocation is dynamic and done at execution time.
 - D) None of the above.

Consider a paging system that supports a 5-bit logical address and has 64 bytes physical memory. The size of a page is 4 bytes.

- (b) (5 points) If the size of a page table entry is 2 bytes (first byte for frame number & second byte for protection bits), then how many bytes are required for a page table?
- (c) (10 points) The following figure shows the physical memory content and the 3rd page table (three-level paging) of a process. Fill the logical content of the process.

Logical address

Logical address	content
0	1
1	3
2	3
3	5
4	8
5	10
6	16
7	15
8	11
9	9
10	9
11	13
12	3
13	4
14	5
15	6
16	7
17	0
18	2
19	5
20	4
21	1
22	7
23	3
24	2
25	4
26	6
27	8
28	13
29	4
30	9
31	9



8	9	2
5	9	2
15	11	3
12	0	4
1	1	5
3	1	6
14	14	7
6	6	

Physical address

Physical address	content
0	1
1	3
2	3
3	5
4	7
5	0
6	2
7	5
8	0
9	3
10	7
11	5
12	4
13	1
14	7
15	3
16	9
17	4
18	11
19	6
20	8
21	10
22	16
23	15
24	13
25	4
26	9
27	9
28	14
29	7
30	6
31	13

8
9
10
11
12
13
14
15

5. Virtual Memory:

interrupt service routine

- (10 points) Describe the actions taken by the operating system when a user process tries to access an invalid memory page. **You need to consider whether the reference is invalid or it is a page fault. You also need to consider page replacement** when there is no free frame for a page fault. In addition, **a modify bit is provided** to improve page-fault service time.
- (10 points) Consider the following logical memory references to a system with 2KB physical memory which is partitioned into 4 frames of the same size. How many page faults would occur for (5 points) the optimal page-replace algorithm and (5 points) the LRU page replacement algorithm?

$$\frac{2KB \cdot 2^{10}}{x} = 4 \quad x = 2^9$$

8

1 001010001001
 2 010011100011
 3 011101110100
 4 100000000111
 2 010111101000
 1 001000111111
 5 101101011111
 6 110111111001
 2 010010000000
 1 001101111000
 2 010111111111
 3 011101111011
 7 111000110000
 6 110100011111
 3 011010100000
 2 010000010111
 1 001000000001
 2 010000010011
 3 011111101101
 6 110010011001

001
 010
 011
 100
 010
 001
 101
 110
 010
 001
 010
 011
 111
 110
 011
 010
 001
 010
 011
 110

1
 2
 3
 4
 2
 1
 5
 6
 2
 1
 2
 3
 7
 6
 3
 2
 1
 2
 3
 6

1 (2)
 2 (4)
 3 (3)
 5 (1)

1 (4)
 2 (2)
 3 (2)
 6 (2)

2 1
 3 1
 6 1
 7 0

1
 2
 3
 6

1
 3
 4
 7
 0
 7
 1
 0
 0
 7
 6
 0

7
 0
 7
 1
 3
 6
 1

2
 1
 2
 6
 3
 7

1
 0
 7

1
 6
 0